

CEREBRUM: Community-Structured Graph Attention for Knowledge Graph Reasoning

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Abstract

CEREBRUM v2.51.0 (Phase 167) represents a paradigm shift in Knowledge Graph reasoning, moving from statistical approximation to deterministic, path-based attention. This manuscript archives the technical specifications, research whitepapers, and empirical benchmarks developed through 167 phases of autonomous engineering.

1 Introduction

CEREBRUM is a framework that enables Knowledge Graphs (KGs) to perform multi-hop reasoning using structural attention heads, replacing the black-box opacity of LLMs with transparent, auditable graph traversal.

2 Theoretical Framework

2.1 Community-Structured Attention (CSA)

CSA provides a global structural constraint on traversal:

$$a(u, v, k) = \sigma(\alpha \cdot \text{sim}(u, v) + \beta \cdot S_{com}(u, v) + \dots) \quad (1)$$

3 Benchmark Performance

CEREBRUM v2.51.0 benchmarks:

- **MetaQA 3-Hop:** 73.2% Hits@10
- **Hetionet:** 85% Hits@10 (Zero-Shot)

4 Conclusion

CEREBRUM provides a blueprint for an autonomous, self-healing knowledge substrate that grows through meaningful connection synthesis rather than brute-force scaling.